

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

ENERGY-BASED MODELS

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- $U(x)$ is the *energy function*
- E.g., $U(x)$ could be a neural network
- Give up on computing the normalization factor!

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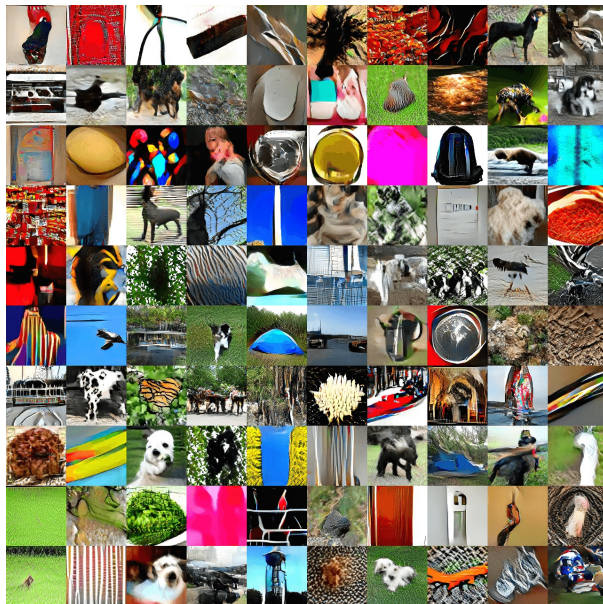
What can we do without normalization?

- Compute likelihoods?
- Sample?
- Generate most likely explanation/completion?
- Learn?

Example

Du, Mordatch (2019). Implicit generation and modeling with energy-based models.

After training on Imagenet128:



Easy to incorporate constraints like **compositionality**.

Sampling from an Energy-based Model

For $p(x) \propto \exp(-U(x))$, can use Gibbs sampling.

Alternative: **Langevin sampler**.

- Initialize $x \in \mathbb{R}^d$
- Repeat:
 - Sample $Z \sim N(0, I_d)$
 - Set $x \leftarrow x - \gamma \nabla_x U(x) + \sqrt{2\gamma} Z$

If $\nabla_x U(x)$ is well-behaved (e.g., smooth), this gets close to $p(\cdot)$.

Learning by Maximum Likelihood

Let $U_\theta(x)$ be the energy function with (e.g., neural net) parameters θ .

$$p_\theta(x) = e^{-U_\theta(x)} / Z_\theta$$

$$Z_\theta = \int e^{-U_\theta(x)} dx$$

Objective: given data $x_1, \dots, x_n$, maximize likelihood.

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Key fact: $\nabla_\theta \ln p_\theta(x) = -\nabla_\theta U_\theta(x) + \mathbb{E}_{x \sim p_\theta}[\nabla_\theta U_\theta(X)]$

- Thus, can use gradient descent
- Estimate $\mathbb{E}_{x \sim p_\theta}[\cdot]$ by sampling from p_θ

We have $p_\theta(x) = e^{-U_\theta(x)}/Z_\theta$ where $Z_\theta = \int e^{-U_\theta(x)} dx$.

Fact: $\nabla_\theta \ln p_\theta(x) = -\nabla_\theta U_\theta(x) + \mathbb{E}_{x \sim p_\theta}[\nabla_\theta U_\theta(X)]$.

First plug in the definition of p_θ to get

$$\nabla_\theta \ln p_\theta(x) = \nabla_\theta (-U_\theta(x) - \ln Z_\theta) = -\nabla_\theta U_\theta(x) - \nabla_\theta \ln Z_\theta$$

So, we just need to focus on the second term. Recall $d(\ln u) = du/u$, and so

$$\begin{aligned} \nabla_\theta \ln Z_\theta &= \frac{\nabla_\theta Z_\theta}{Z_\theta} = \frac{\nabla_\theta \int e^{-U_\theta(x)} dx}{Z_\theta} \\ &= \frac{\int \nabla_\theta e^{-U_\theta(x)} dx}{Z_\theta} = \frac{-\int e^{-U_\theta(x)} \nabla_\theta U_\theta(x) dx}{Z_\theta} \\ &= -\int p_\theta(x) \nabla_\theta U_\theta(x) dx = -\mathbb{E}_{X \sim p_\theta}[\nabla_\theta U_\theta X]. \end{aligned}$$